

Study 2 Writeup

Feature generation key takeaways

We use two approaches for generating features:

1. **BERTopic:** Clusters post embeddings and generates topics based on the substance of the text.
2. **LLM-driven feature generation:** generates features based on both the substance of text and the civility of conversation.

The key takeaway across both these methods is that:

- The topic, substance, or political lean of a post are weak predictors of keep/remove behavior
- The civility of conversation (i.e., the toxicity of language) is a strong predictor.

BERTopic on the original posts

We focus the BERTopic analysis on the original posts. The mirroring operation did change some of the topics from the original post. We try to keep the same stance and political intensity, but there are different topics that are triggers for left-leaning and right-leaning groups. We add a follow-up set of analyses later on reviewing how the mirroring approach changed the topics. We also use all keep/remove decisions per post, rather than taking a majority vote. For convention's sake, we measure the overall keep rate, rather than the remove rate.

We see more well-defined BERTopic results from the larger combined Study 2 dataset than we did in the first run given our larger dataset and the increased number of labels per post.

Overview

Below is a 2-D visualization of the posts clustered by their key topics. An interactive visualization can be found at [this link](#).

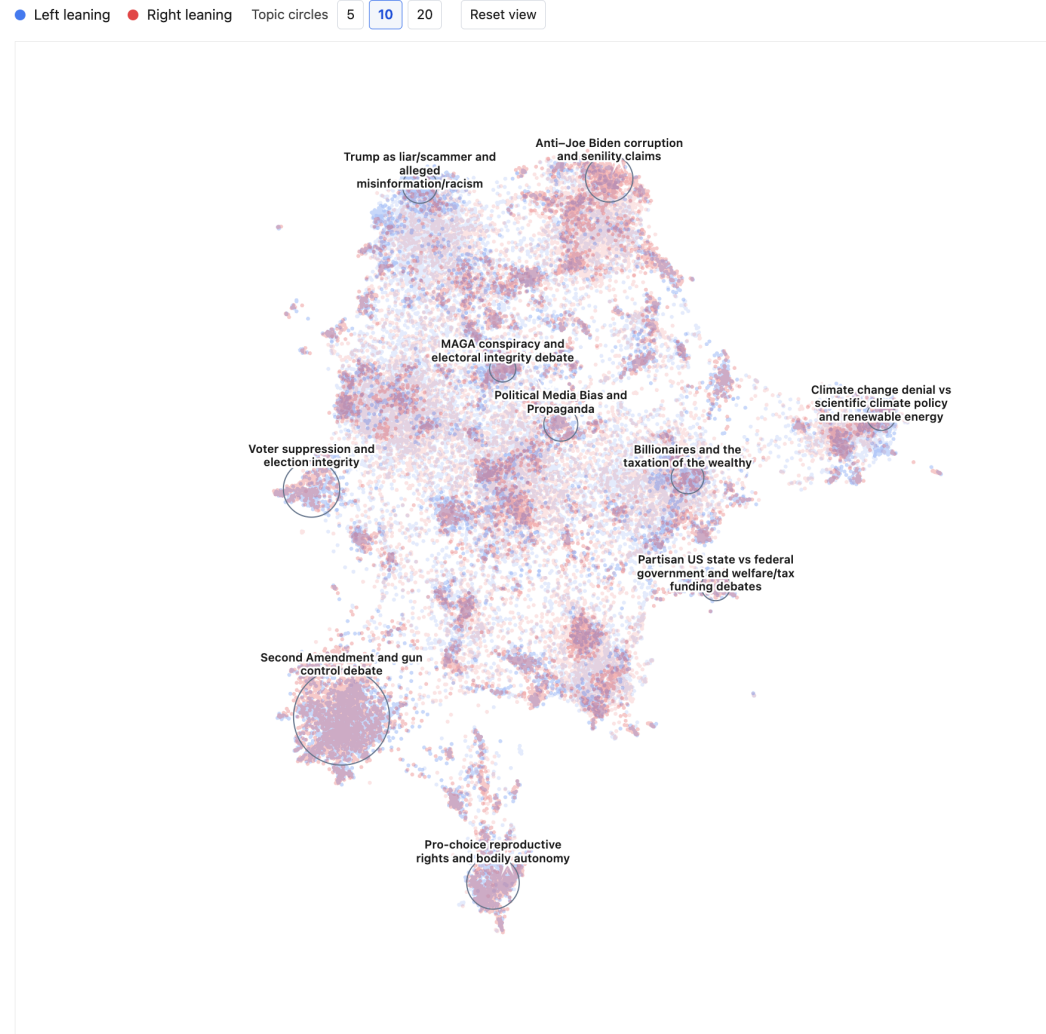


Figure 1: BERTopic Cluster Map

Which topics most commonly appear?

Topic	Posts	Keep rate
Climate and fossil fuels	1,122	81.8%
Border and immigration	862	75.8%
MAGA and institutions	851	59.1%
Abortion access	690	75.2%
Trump media and lying	636	59.3%
Criticism of Democrats	598	69.3%

Topic	Posts	Keep rate
Criticism of Republicans	566	59.0%
Biden and Trump blame	358	67.6%
Billionaires and taxes	326	76.6%
Abolish ICE	259	62.1%

Which topics are more common in originally left leaning posts, and which in right leaning posts?

Each post and its mirror are shown, so users see both the left-leaning and right-leaning versions. We analyze, based on the original posts, which topics tend to be more common in left-leaning vs. right-leaning posts.

Some topics appear much more often in left- or right-leaning posts

Share of posts within each political-leaning group, by topic

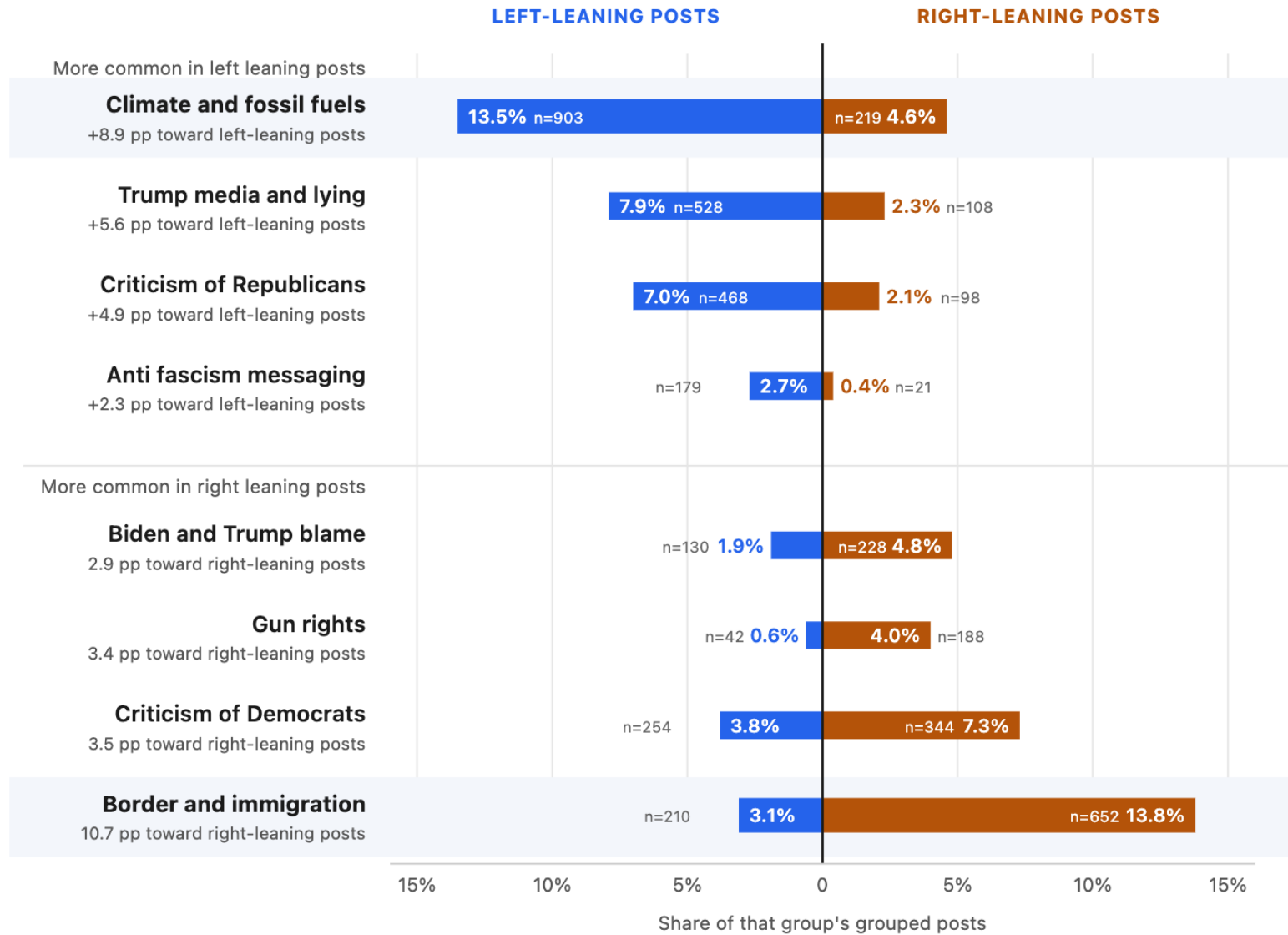


Figure 2: Topics by Left/Right-Leaning

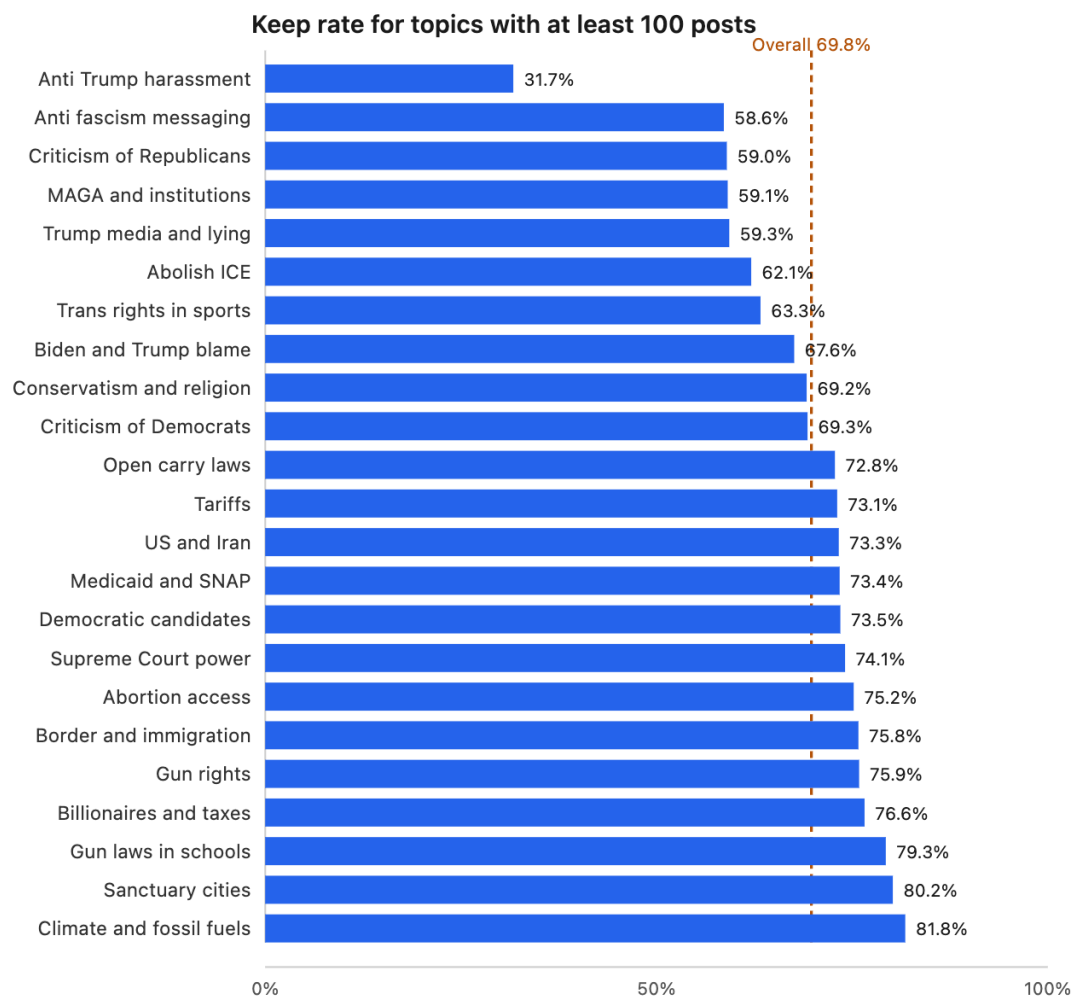
Which topics are more common at low, medium, and high toxicity?

Low toxicity posts are more often about climate, abortion, and billionaires and taxes. Climate alone is 20.5% of low toxicity grouped posts and 1.6% of high toxicity grouped posts. High toxicity posts are more often attacks on Trump, MAGA and institutions, and criticism of Republicans. Criticism of Trump’s media behavior is 13.9% of high toxicity grouped posts and 1.1% of low toxicity grouped posts. Anti Trump harassment is almost entirely in the high toxicity sample (140 of 144 posts).

Topic	Low	Medium	High
Climate and fossil fuels	20.5% (604)	8.5% (471)	1.6% (47)
Abortion access	8.2% (240)	7.0% (389)	2.1% (61)
Billionaires and taxes	4.8% (141)	2.1% (117)	2.4% (68)
Trump media and lying	1.1% (33)	3.6% (200)	13.9% (403)
Anti Trump harassment	0% (0)	0.1% (4)	4.8% (140)
MAGA and institutions	3.3% (98)	7.9% (439)	10.9% (314)
Criticism of Republicans	2.2% (66)	4.9% (272)	7.9% (228)

Do some topics get removed more often?

Among the 23 topics with at least 100 posts, the keep rate runs from 31.7% to 81.8%. The topics most commonly removed include anti-Trump harassment and anti-fascism messaging, while the topics most commonly kept include climate change and sanctuary cities.



Each bar is one topic with at least 100 posts. The dashed line is the overall keep rate, 69.8%. The percentage is the average share of raters who voted keep.

Figure 3: Keep Rate by Topic

Which topics did Democratic/Republican raters keep, and which did they remove?

Democratic raters kept 69.4% of 52,864 votes. Republican raters kept 70.0% of 48,970 votes. The topics they were most likely to remove, and the topics they were most likely to keep, are almost the same list. Both groups removed anti Trump harassment most often, at about 32%. Both groups kept climate and fossil fuels most often among large topics, at about 82%.

Where the two groups differ, the gap is generally small. Across 26 topics with at least 200 votes from each group, the typical gap is under 2 points. The largest gaps are:

- Open carry laws: Democratic raters kept 75.8% of 289 votes, and Republican raters kept 68.3% of 249 votes.
- Supreme Court: Republican raters kept Supreme Court posts at 77.6% (308 votes), and Democratic raters kept them at 71.5% (330 votes).

Keep/remove behavior for Democrats Here are the topics Democratic raters were most likely to remove, among topics with at least 200 of their votes.

Topic	Keep rate	Votes
Anti Trump harassment	31.5%	444
MAGA and institutions	57.6%	2,308
Criticism of Republicans	59.1%	1,500
Trump media and lying	59.4%	1,788
Anti fascism messaging	59.7%	571

Here are the topics Democrat raters were most likely to keep:

Topic	Keep rate	Votes
Gun policy debate	82.4%	262
Climate and fossil fuels	82.2%	2,872
Gun laws in schools	79.5%	410
Sanctuary cities	78.6%	266
Gun rights	77.6%	602

Keep/remove behavior for Republicans Here are the topics Republican raters were most likely to remove, among topics with at least 200 of their votes.

Topic	Keep rate	Votes
Anti Trump harassment	31.9%	282
Anti fascism messaging	57.7%	478
Criticism of Republicans	59.1%	1,377
Trump media and lying	59.2%	1,473
MAGA and institutions	60.8%	2,059

Here are the topics Republican raters were most likely to keep:

Topic	Keep rate	Votes
Climate and fossil fuels	81.5%	2,893
Sanctuary cities	80.6%	289
Gun policy debate	80.2%	242
Gun laws in schools	79.1%	369
Mail and elections	78.2%	257

Does the stance of the post affect the keep/remove rate? The stance of the post barely moves the overall keep rate. Left leaning posts are kept at 69.0% (11,386 posts). Right leaning posts are kept at 70.9% (8,377 posts). Within topics, certain topics present in left-leaning or right-leaning posts are kept more often:

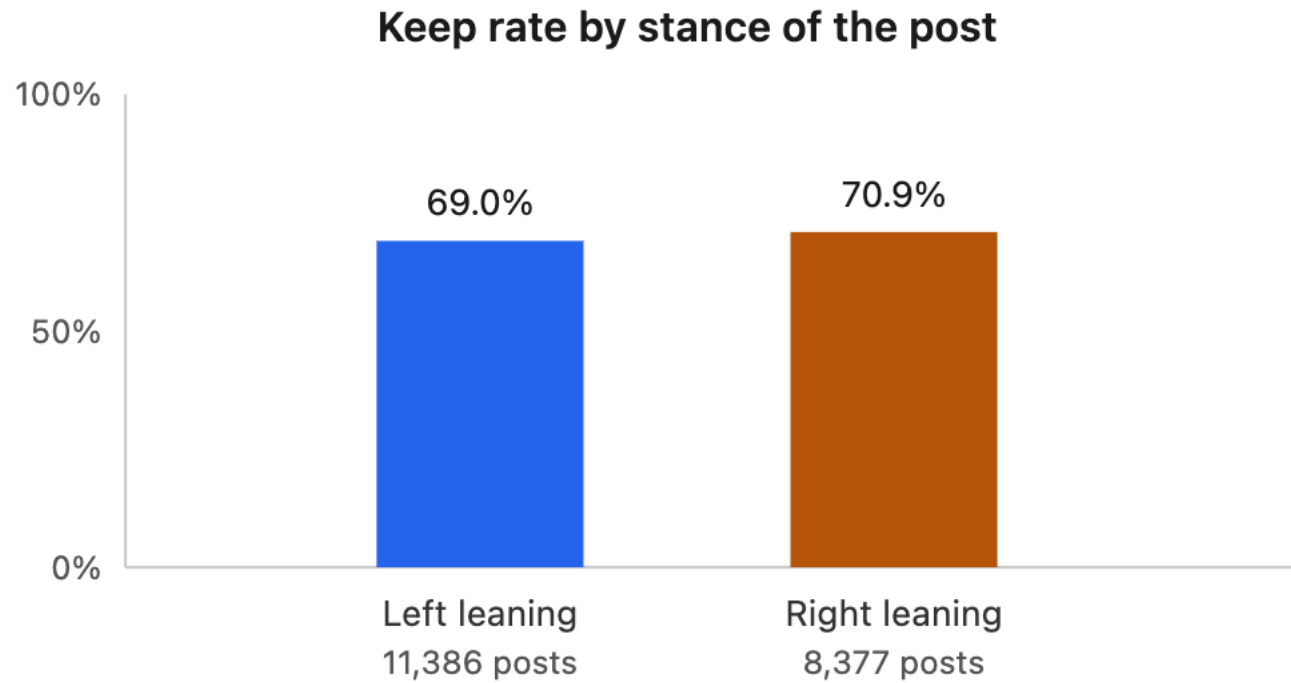


Figure 4: Keep Rate by Political Lean

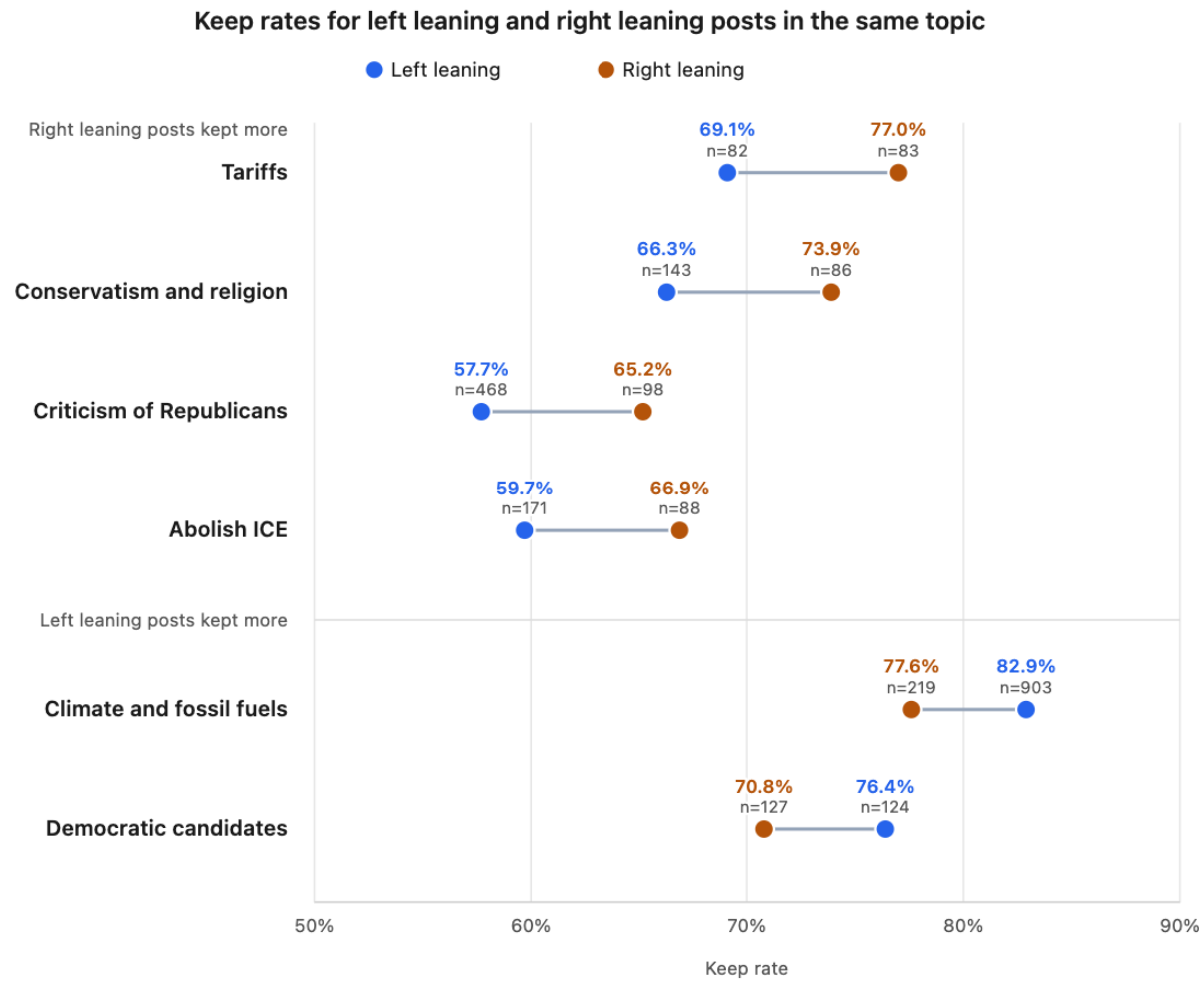


Figure 5: Keep Rate by Topic and Political Lean

Does the toxicity of the post affect the keep/remove rate? Toxicity moves the keep rate much more than stance does. Low toxicity posts are kept at 84.7% (4,892 posts). Medium toxicity posts are kept at 72.6% (9,888 posts). High toxicity posts are kept at 49.8% (4,983 posts).

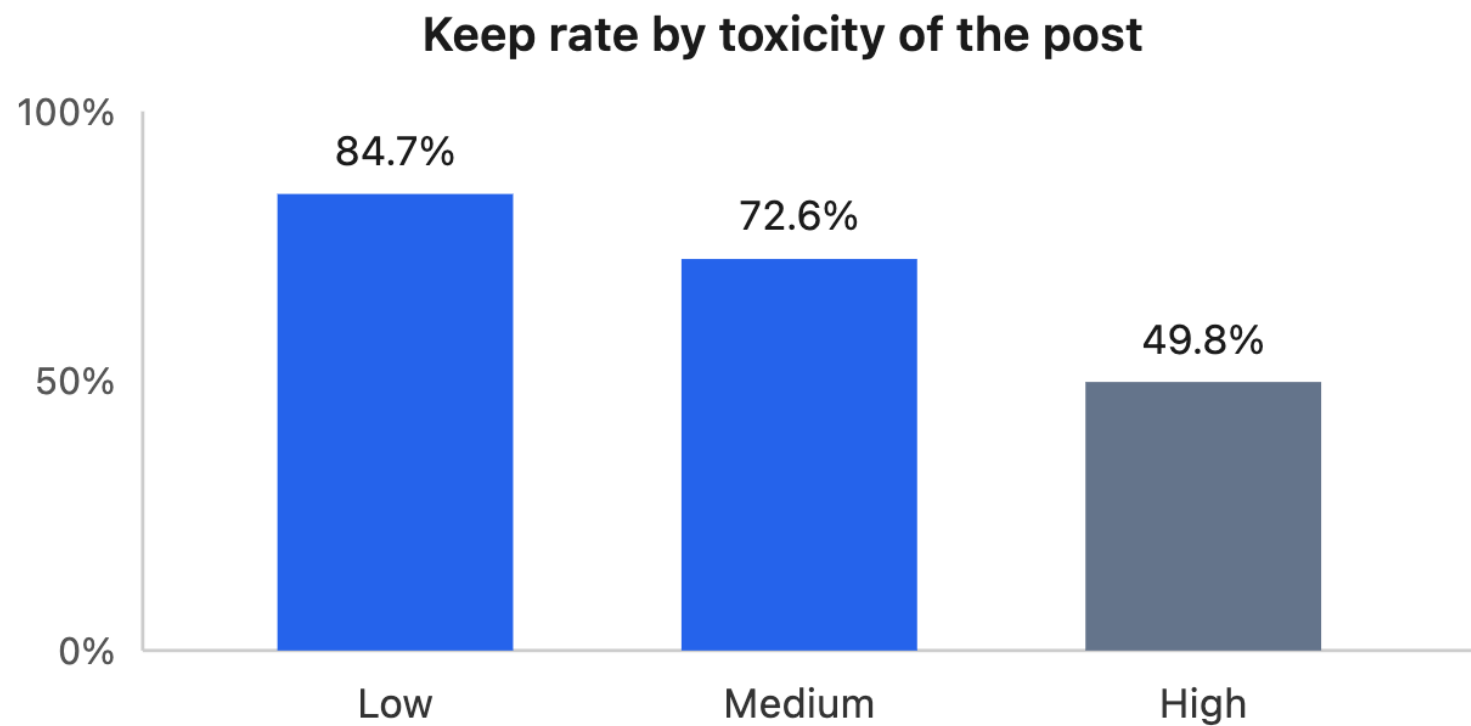


Figure 6: Keep Rate by Toxicity

This trend is generally true across topics as well:

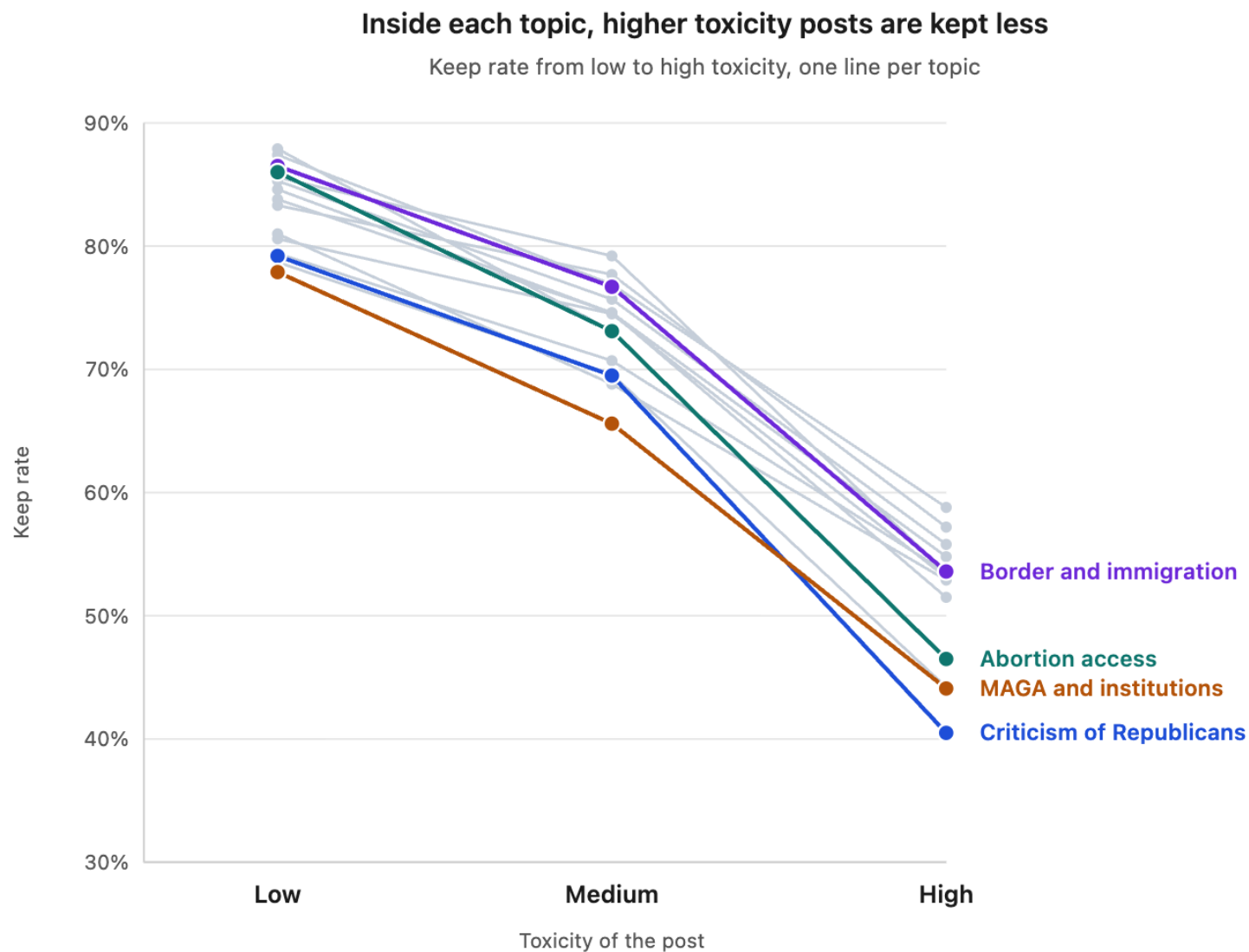


Figure 7: Keep Rate by Toxicity and Topic

LLM-based feature generation

We then use LLMs to perform feature extraction. We follow a protocol developed by Google DeepMind on how to use LLMs for automated feature extraction.

Methods

1. **Setup:** We assign a keep/remove label to each original + mirror combination based on the modal label.

2. **Use an LLM to mine features:** We pass in batches of 10 pairs of posts that were majority keep and 10 pairs of posts that were majority remove. We then ask an LLM to extract features to distinguish posts that were kept and posts that were removed. We do this across a few categories of features (see below).
3. **Embed the feature records and cluster them:** We embed the features and then use K-Means to generate clusters. We do this instead of recursively asking an LLM to generate a feature category given a list of features since we want a sense of “global similarity” across all features. We choose K-Means since HDBSCAN rendered unstable estimates.
4. **Name each cluster:** We take each cluster and pass them to an LLM to generate a human-readable label and description of each feature cluster. This generates a compiled list of feature groups.
5. **Label every post against the features:** We label each post against each feature group. This means that a post can have multiple features, which makes this approach more feature-rich and expressive than BERTopic.

Categories of features

These are the categories that we gave to the LLM when asking them to mine for features in the original posts:

1. **Surface and lexical:** This category is about how the post is written, not what claim it makes. It covers length, slang, heavy punctuation, all-caps emphasis, profanity, hashtags and account mentions, and a high density of proper names. Examples include emphatic typography, profane derogatory insults, colloquial language and insults, and hashtags and account mentions.
2. **Topic and subject matter:** This category is about the subject of the post. Examples include a policy area (guns, climate, immigration, abortion, elections), a specific event or bill, a geographic scope, a historical analogy, and culture-war salience.
3. **Semantic content:** This category is about the kind of claim the post makes. Examples include causal claims, moral language, a factual claim versus speculation, conspiracy, a claim that a group is being persecuted, a policy prescription, and a cost-benefit argument.
4. **Pragmatics and communicative intent:** This category is about what the post is doing to the reader. Examples include sarcasm, mockery, a call to action, persuasion, venting, hedging, and outrage.
5. **Target and directionality:** This category is about who the post attacks or praises, and which political side it points at. Examples include the type of actor criticized or praised, a left/right cue, us-versus-them framing, and elite-versus-populist framing.
6. **Compositional and syntactic structure:** This category is about the shape of the sentences. Examples include if-then conditionals, contrast with “but” or “however,” rhetorical questions, parallel repetition, lists, quoted or attributed speech, and direct address in the second person.

Discovered categories

After mining features, embedding, and naming each resulting feature cluster, here are the feature categories that were discovered:

Name	Definition
Blanket Demonization of Political Opponents	Sweeping, unqualified portrayals of an opposing political group as inherently stupid, immoral, corrupt, dangerous, subhuman, or traitorous.
Brief Slogan-Like Insult Attacks	Pairs feature very short, blunt, insult-led statements or fragments that make a denunciatory claim with little or no supporting elaboration.
Broad Partisan Out-Group Targeting	The pair targets or refers to ordinary members, supporters, voters, or whole populations defined by broad partisan or ideological identities rather than only specific leaders, institutions, or policies.
Culture-War Policy Issues	Posts focus on contested partisan policy debates over guns, immigration and border enforcement, abortion and reproductive rights, policing, or related rights-based social issues.
Culture-War Security and Out-Group Threats	Posts frame immigration, policing, guns, identity issues, religion, or political unrest as existential threats to public safety, national identity, or social order.
Dense Profanity and Vulgar Insults	The pair contains frequent or repeated explicit profanity, obscenity, vulgarity, or derogatory insult terms directed at people or groups.
Derogatory Labels and Epithets	The post pair uses insulting, demeaning, slur-like, or dehumanizing labels to characterize people or political groups.

Name	Definition
Direct Second-Person Hostility	Posts directly address an individual or group as “you,” “you guys,” or similar terms while using insults, taunts, accusations, threats, or hostile commands.
Electoral Strategy and Political Consequences	Posts that predict, warn about, or explain how parties, politicians, voters, policies, or messaging will affect electoral outcomes, public support, political behavior, or longer-term political consequences.
Elite-versus-Public Political Framing	Frames political or economic conflict as powerful elites, institutions, or wealthy interests opposing ordinary people, workers, voters, or taxpayers.
Escalatory Partisan Hostility	Posts use inflammatory us-versus-them rhetoric, contempt, threats, catastrophe claims, or gloating to intensify hostility toward political opponents rather than argue policy.
Explicit Contrastive Framing	The pair uses overt contrast markers or alternative constructions—such as “but,” “instead,” “while,” “rather than,” or “not X, but Y”—to set competing claims, actions, or groups against each other.
Extended Explanatory Argumentation	Posts develop a claim through multiple sentences or paragraphs that provide explanation, reasons, supporting details, examples, or justification rather than relying solely on a brief slogan or assertion.
Hostile Outrage Venting	Posts primarily express anger or outrage through hostile denunciation, contempt, ridicule, insults, or profanity toward a target.
Limited Profanity Within Argument	The pair uses occasional mild profanity, slang, or insults within broader substantive commentary rather than sustained, standalone, or slur-heavy abuse.
Opponent Claims Framed for Rebuttal	The post quotes, paraphrases, or rhetorically challenges an opposing claim, slogan, or premise and then argues against or undermines it.
Partisan Collective Blame	Attributes broad social, political, economic, or violent harms to an entire political party, ideology, supporter base, or other large partisan out-group rather than to specific individuals or actions.
Partisan Mockery and Taunting	Posts use sarcasm, ridicule, caricature, mock quotations, or taunting rhetorical questions to demean, humiliate, or belittle political opponents.
Personal Attacks on Political Figures	Posts directly target identifiable politicians, public officials, or associated supporters with hostile personal insults, character attacks, or derogatory labels.
Policy Advocacy and Tradeoff Arguments	Posts advocate, oppose, or critique specific government policies or reforms, often explaining their expected consequences, tradeoffs, implementation, or priorities.
Policy Consequence Warnings and Civic Mobilization	Posts express concern about governmental, institutional, or policy harms by citing consequences for rights, safety, fairness, or material well-being and often urge accountability, reform, voting, or other political action.
Political Actor and Institution Criticism	Posts criticize political parties, leaders, ideological groups, or institutions for their policies, conduct, competence, corruption, or governing choices.
Political and Institutional Targets	The pair primarily targets political parties, elected officials, ideological factions, government bodies, or related elite institutions and organized interests.
Punitive Blanket Condemnation of Political Opponents	Posts broadly demonize political opponents or their supporter groups and advocate or wish for their punishment, exclusion, removal, or other punitive treatment.
Quoted Slogans and Emphatic Framing	Posts use quoted or slogan-like political language, sometimes with capitalization, scare quotes, italics, or exclamation marks, to frame, attribute, critique, or comment on claims.
Reasoned Political Policy Persuasion	Posts seek to persuade through substantive political or policy arguments, using explanations, justifications, evidence, examples, or rebuttals rather than merely asserting a position.
Shouting-Style Emphatic Formatting	Posts use conspicuous all-caps, repeated exclamation or other emphatic punctuation, slogans, fragments, repetition, or intensifiers to convey a loud, urgent, confrontational tone.
Specific Political and Institutional References	Posts use concrete names of politicians, parties, agencies, institutions, policies, legal concepts, or issue-specific political terminology rather than only broad political language.
Substantive Policy and Institutional Claims	Posts make concrete factual, causal, or interpretive claims about laws, government actions, institutions, political actors, or their social, economic, and rights-related consequences.

Sweeping Unsubstantiated
Political Accusations

Posts make broad, categorical allegations that political opponents or leaders are corrupt, criminal, authoritarian, immoral, dishonest, or otherwise malign without qualifying evidence or nuance.

Results

We have evidence that suggests that people moderate on the civility of conversation and less on the substance of the conversation.

We find that the probability of a post being removed is correlated to increases in hostile language and negatively linked to material policy arguments. Substance-related references and mentions of political actors remain relatively constant.

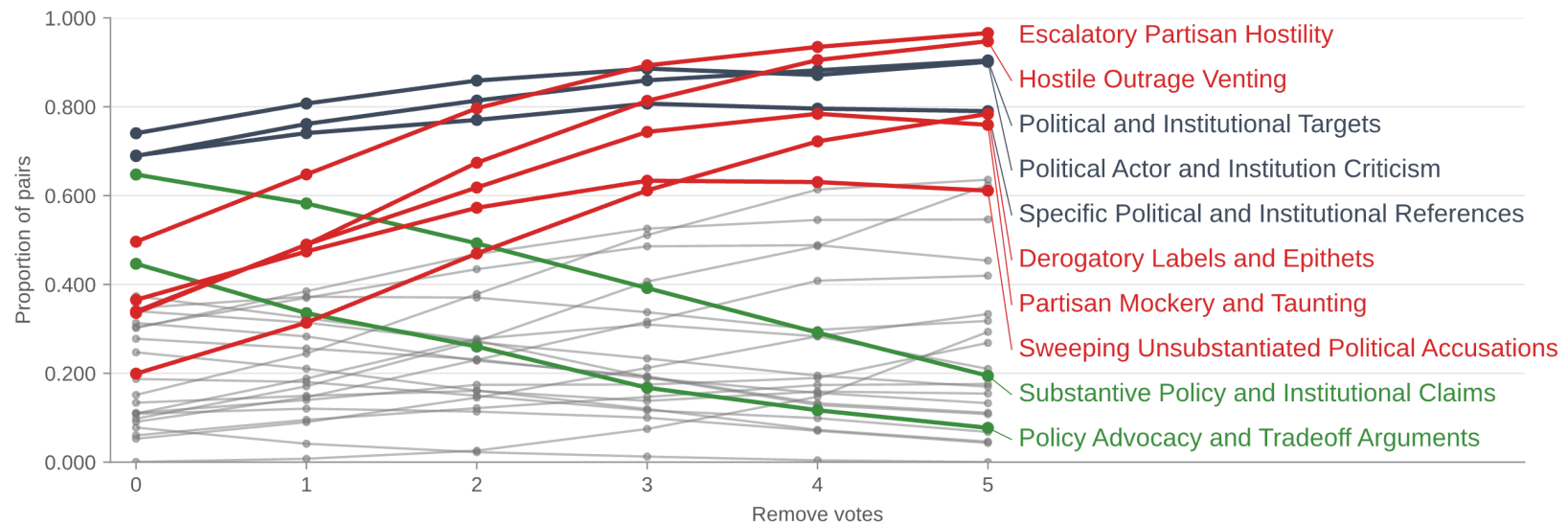


Figure 8: Topic Distribution by Keep/Remove

Similarly, we find that as toxicity of the post increases, hostile language increases and policy argument decreases, while substance-related references and mentions of political actors remain relatively constant.

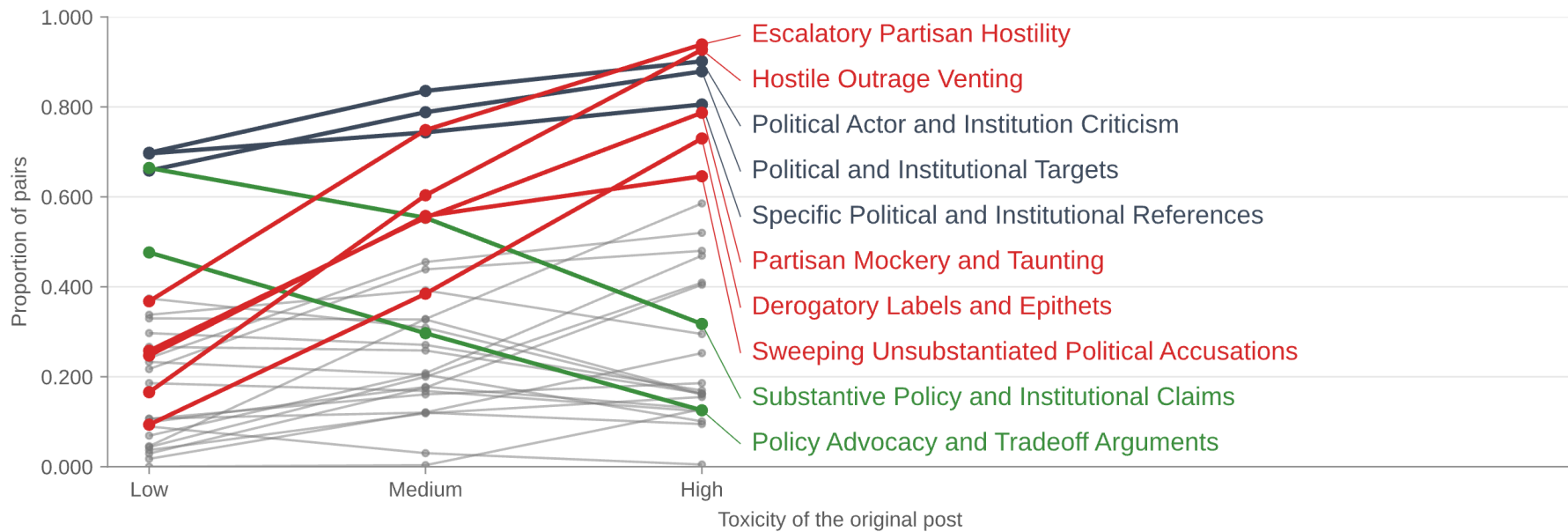


Figure 9: Topic Distribution by Toxicity

Training a binary classifier

Work is still ongoing and pending results

We train a variety of binary classifiers to predict the keep/remove task. We experimented with the following approaches:

1. Zero-shot LLM inference
2. Few-shot LLM inference
3. Few-shot prompt-tuned LLM inference
4. Fine-tuning an open-source LLM

We test across the following models:

- Amazon Nova
- Qwen 3 32B
- OpenAI GPT-5.6 Terra
- Claude Sonnet 5.5

For fine-tuning, we use Qwen3.8-27B. We deploy using vLLM and we use a quantized deployment.

Dataset

We evaluate our models across 3 versions of our dataset. We filter for posts with exactly 5 labels, to avoid ties in labels (n=13,992):

1. Unanimous posts (n=4,051): posts that were unanimously kept or unanimously removed across labelers.
2. Split posts (n=9,941): posts that were neither unanimously kept nor unanimously removed across labelers.

3. All posts (n=13,992): the total dataset of posts. For these posts, we take the modal label.

Prompts

Zero-shot prompt We use the same prompt that was shown to study participants, randomly shuffling the order of the original and mirror post:

We are developing a new social media platform designed specifically for political discussion. As part of this process, we'd like to understand how

The pairs are political mirrors of each other. This means that the mirror text recreates the original message from the opposite political stance.

Original Text:

I'm a bleeding-heart liberal, and I think the issue of abortion is obviously about protecting women's rights!

Mirror Text:

I'm a staunch conservative, and abortion is fully about the sanctity of human life before birth!

Notice that the mirror text changes the core message to match that of the opposite political stance. In other words, the mirror text is not a resp

Your job is to decide whether both posts in the pair should be allowed or removed from the platform.

When making your decisions, consider generally whether a post contributes to a healthy environment for political discussion, or whether it would b

Post 1: {post_1_text}

Post 2: {post_2_text}

Allow Or Remove?

Baselines

Zero-shot baselines Across all posts:

Model	F1	Accuracy	Recall	Precision
Amazon Nova Micro	0.426029	0.550529	0.786388	0.292152
Qwen 3 32B	0.498773	0.649800	0.821429	0.358108
OpenAI GPT-5.6 Terra	0.316304	0.800100	0.217992	0.576135
Claude Sonnet 5.5	0.223602	0.803459	0.133423	0.689895
Jev	0.529322	0.712621	0.761792	0.405561

Across unanimous posts:

Model	F1	Accuracy	Recall	Precision
Amazon Nova Micro	0.256932	0.603061	0.902597	0.149784

Model	F1	Accuracy	Recall	Precision
Qwen 3 32B	0.395946	0.779314	0.951299	0.250000
OpenAI GPT-5.6 Terra	0.478528	0.937053	0.379870	0.646409
Claude Sonnet 5.5	0.382134	0.938534	0.250000	0.810526
Jev	0.493617	0.853123	0.941558	0.334487

Across split posts:

Model	F1	Accuracy	Recall	Precision
Amazon Nova Micro	0.467645	0.529122	0.772932	0.335236
Qwen 3 32B	0.517117	0.597022	0.806391	0.380589
OpenAI GPT-5.6 Terra	0.294281	0.744291	0.199248	0.562633
Claude Sonnet 5.5	0.203249	0.748416	0.119925	0.665971
Jev	0.535016	0.655367	0.740977	0.418649

Few-shot baselines We use a similar prompt as the zero-shot version, but we append 5 few-shot examples of post pairs that were kept and 5 few-shot examples of post pairs that were removed. We sampled posts that were either unanimously kept or unanimously removed, and model training and evaluations exclude these few-shot samples.

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Your job is to decide whether both posts in the pair should be allowed or removed from the platform.

When making your decisions, consider generally whether a post contributes to a healthy environment for political discussion, or whether it would be harmful

Post 1: {post_1_text}

Post 2: {post_2_text}

Here are examples of pairs of posts that human annotators remove:

1. Post 1: Conservative Christians are on the side of the capitalist class or billionaires and should be physically fought. #communism #socialism

2. Post 1: Fuck this smug DNC-groomed hypocrite. The day he actually fights for the Second Amendment will be the first day, since he's been kissin
3. Post 1: Woke, DEI, social justice, critical race theory, politically correct, civil rights, restorative justice, reparations... Just use the ha
4. Post 1: And why there are far fewer soy-brained libtard town hall meetings to 'splain why Biden's corruption and cognitive decline is good for
5. Post 1: Democrats will never be an honest decent party with integrity, generosity, and warmth - it's all about grasping for power and quite a b

Here are examples of pairs of posts that human annotators keep:

1. Post 1: If everyone who says something to the effect of "gun rights are important for hunters" understood how stupid the average Wisconsinite h
2. Post 1: For decades the US left told us we shouldn't impose American values on other countries - sovereignty matters, they said. But now they're falling all over themselves to copy Europe's socialist failures: crushing taxes, open
3. Post 1: always under siege Second Amendment. Lloyd Smucker has my Complete and Total Endorsement for Re-Election. Election Day is Tuesday, May
4. Post 1: The Secretary of State should be charged with voter suppression by making it harder for working people to cast their ballots in person.
5. Post 1: I take exception to some of how the definition of "assault weapon" is formed. New York, where I grew up, made significant regulations b

Allow Or Remove?

Automated prompt tuning

We performed automated prompt tuning using DSPy and the GEPA optimization metric (CITE).

Training a calibrated classifier

Work is still ongoing and pending results

One of the shortcomings of building a binary classifier is that it imposes a discrete label on an inherently uncertain task. In a non-trivial amount of tasks, people themselves were uncertain of what the label should have been. We turn this into the following prediction task:

$p(\text{label}) = p(\text{proportion of 5 raters that would, on average, remove this post})$

We present several approaches for this:

1. **Few-shot prompt-tuned LLMs that return their own probabilities:** We ask LLMs to report their own probabilities.
2. **Few-shot prompt-tuned calibrated classifier (Jev):** We prompt-tune Jev using DSPy and GEPA.
3. **An ensemble approach.** (Add more details)
4. **A calibrated classifier.** We can develop a classifier that returns a probability, and we can then set an arbitrary threshold to generate keep/remove decisions. This is inspired by how the Perspective API was built and follows past work on reinforcement learning with calibration rewards (RLCR).

Prompt-tuning a calibrated classifier

One of the shortcomings of building a binary classifier is that it imposes a discrete label on an inherently uncertain task. In a non-trivial amount of tasks, people themselves were uncertain of what the label should have been. There's not much clarity on this

Jev + DSPy + GEPA.

Fine-tuning a calibrated classifier

Qwen.

Calibrated classifier fine-tuning results

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